

2026 MIDDLE SCHOOL STEM CAMPS

# From Trees to Tech

## INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

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# MudWatt Bioelectric League

Turn a cup of mud into a living battery

**At a glance:** Can your team make mud generate electricity and defend your design with data? About 80 min; 4 teams.

## MATERIALS

- 4 DIY soil microbial fuel cells: clear deli container, pond mud, two 4×4 in carbon-felt electrodes each (plus 1 spare set)
- measured sugar cubes as fuel (one per cell per run, plus spares)
- 4 multimeters, one per team, plus 2 spare (confirm battery type and that a battery is included)
- 100 kΩ resistors and alligator-clip lead wires, one set per cell
- nitrile gloves and a containment tray per station
- labels and masking tape
- handwashing supplies

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which design change moved your readings the most, and how do you know?
- Why must the anode stay buried and the cathode stay in air?
- How is your weekly log like the work of an energy engineer?

## SCORING (100 POINTS)

| SCORING CRITERION            | POINTS     |
|------------------------------|------------|
| Peak voltage reading         | 35         |
| Daily data log quality       | 20         |
| Controlled-variable design   | 15         |
| Redesign or maintenance plan | 15         |
| Final explanation            | 15         |
| <b>Total</b>                 | <b>100</b> |

**SOURCES** soil microbial fuel cell electrode and voltage references (carbon-felt anodes; terrestrial MFC literature on 300 to 600 mV output as the biofilm matures).

**PRINT: TTT-01 DAILY VOLTAGE LOG**

One per team, on cardstock (or a sheet protector for dry-erase). Each day, set the multimeter to DC millivolts and read the voltage across the 100 kΩ resistor at the same time; the cell is weak on day 1 and climbs as the biofilm grows. Defend the design with the trend, not one number.

**Daily voltage log**

Team \_\_\_\_\_ The ONE variable we are testing \_\_\_\_\_

| DAY          | TIME | VOLTAGE (mV) | NOTES |
|--------------|------|--------------|-------|
| Mon Jun 22   |      |              |       |
| Tue Jun 23   |      |              |       |
| Wed Jun 24   |      |              |       |
| Thu Jun 25   |      |              |       |
| Fri Jun 26   |      |              |       |
| Peak reading |      | Day of peak  |       |

# Forest Sensor Sprint

Site the smartest place for a campus tree sensor

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**At a glance:** Find the smartest place to install a future campus tree sensor. About 80 min; 4 teams.

## MATERIALS

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- 4 clipboards plus spares
- 4 paper clinometers plus spares
- 4 combined temperature + humidity meters (one device, plus spares)
- 4 three-in-one soil meters (moisture, light, pH), one per team, plus spares
  - 1 to 2 long tape measures (pre-mark a fixed standoff distance in prep)
- 4 route cards (one per team) plus 2 spares
- pencils

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

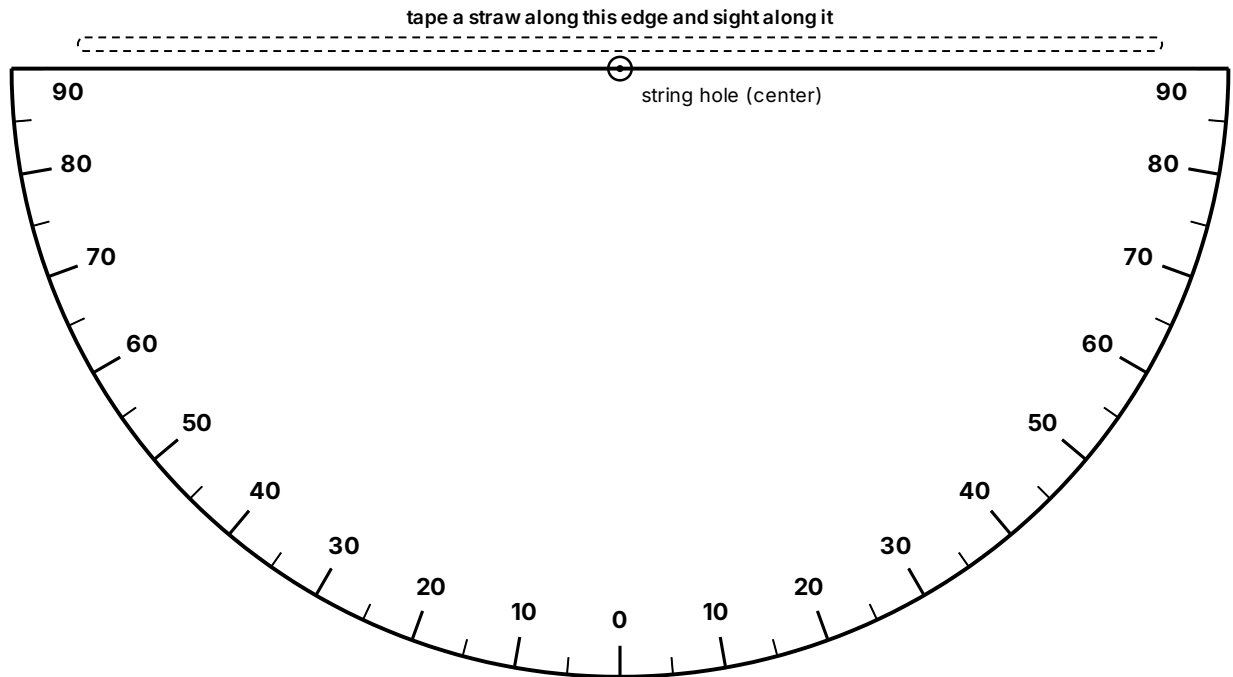
- Which measurement most influenced your choice?
- Where did two nearby spots disagree, and why?
- What would change your recommendation in winter?

**SCORING (100 POINTS)**

| SCORING CRITERION            | POINTS     |
|------------------------------|------------|
| Data accuracy                | 30         |
| Route efficiency             | 20         |
| Site recommendation evidence | 30         |
| Teamwork and field conduct   | 20         |
| <b>Total</b>                 | <b>100</b> |

**SOURCES** GLOBE Observer Trees; NASA MyNASAData Urban Heat Island

# Paper clinometer



Cut along the flat top edge and the curved outline.

## ASSEMBLE

- 01 Cut out the half circle along the curved outline.
- 02 Tape a straight straw along the top straight edge.
- 03 Punch the center dot, thread a string through it, and tie a washer or paper clip on the end as a weight.
- 04 Sight the treetop through the straw, let the string hang free, then pinch it against the scale and read the number.
- 05 Self-check: sight something level (reads 0) and straight up (reads 90).

## FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor marked.

| ANGLE | TAN  |
|-------|------|
| 10    | 0.18 |
| 20    | 0.36 |
| 30    | 0.58 |
| 40    | 0.84 |
| 45    | 1.00 |
| 50    | 1.19 |
| 60    | 1.73 |
| 70    | 2.75 |

Example: eye height 1.5 m, distance 10 m, angle 40° gives  $1.5 + 10 \times 0.84 =$  about 9.9 m.

# Team route card

## From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team \_\_\_\_\_ Date \_\_\_\_\_

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

| CHECKPOINT            | VISIT ORDER | TEMP (°F) | HUMIDITY (%) | LIGHT (RELATIVE) | SOIL MOISTURE (1 TO 10) | NOTES |
|-----------------------|-------------|-----------|--------------|------------------|-------------------------|-------|
| Reference (calibrate) |             |           |              |                  |                         |       |
| A                     |             |           |              |                  |                         |       |
| B                     |             |           |              |                  |                         |       |
| C                     |             |           |              |                  |                         |       |
| D                     |             |           |              |                  |                         |       |
| E                     |             |           |              |                  |                         |       |

Our recommended sensor site: \_\_\_\_\_

Why (use your data): \_\_\_\_\_

The checkpoints are the spots the instructor marked outside.

cut here

## From Trees to Tech · TTT-02 Forest Sensor Sprint · Team Route Card

Team \_\_\_\_\_ Date \_\_\_\_\_

Plan an order that avoids backtracking, take a reading at each checkpoint, then recommend the best sensor site using your numbers.

| CHECKPOINT            | VISIT ORDER | TEMP (°F) | HUMIDITY (%) | LIGHT (RELATIVE) | SOIL MOISTURE (1 TO 10) | NOTES |
|-----------------------|-------------|-----------|--------------|------------------|-------------------------|-------|
| Reference (calibrate) |             |           |              |                  |                         |       |
| A                     |             |           |              |                  |                         |       |
| B                     |             |           |              |                  |                         |       |
| C                     |             |           |              |                  |                         |       |
| D                     |             |           |              |                  |                         |       |
| E                     |             |           |              |                  |                         |       |

Our recommended sensor site: \_\_\_\_\_

Why (use your data): \_\_\_\_\_

The checkpoints are the spots the instructor marked outside.



One per tree, on cardstock (lamine; it sits on the ground). Place it at the distance you pre-mark from the tree base with the long tape.

FROM TREES TO TECH 2026 • TTT-02

*Tree-height station*

# Stand here

*Sight the treetop through your clinometer  
straw from this mark.*

This mark is a measured distance from the tree base.  
Write that distance here: \_\_\_\_\_ m, then use it in  
your tree-height formula.

# Seed Dispersal Derby

Engineer a seed that escapes its parent tree

**At a glance:** Build a seed that escapes the parent tree without engines or throwing. About 80 min; 4 teams.

## MATERIALS

- paper
- cardstock
- paper clips
- masking tape
- string
- fan or marked test lane
- a shared tape measure for distance
- timers

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which change improved hang time without losing distance?
- Why does spinning slow the fall?
- Which tree's strategy did your final seed copy?

## SCORING (100 POINTS)

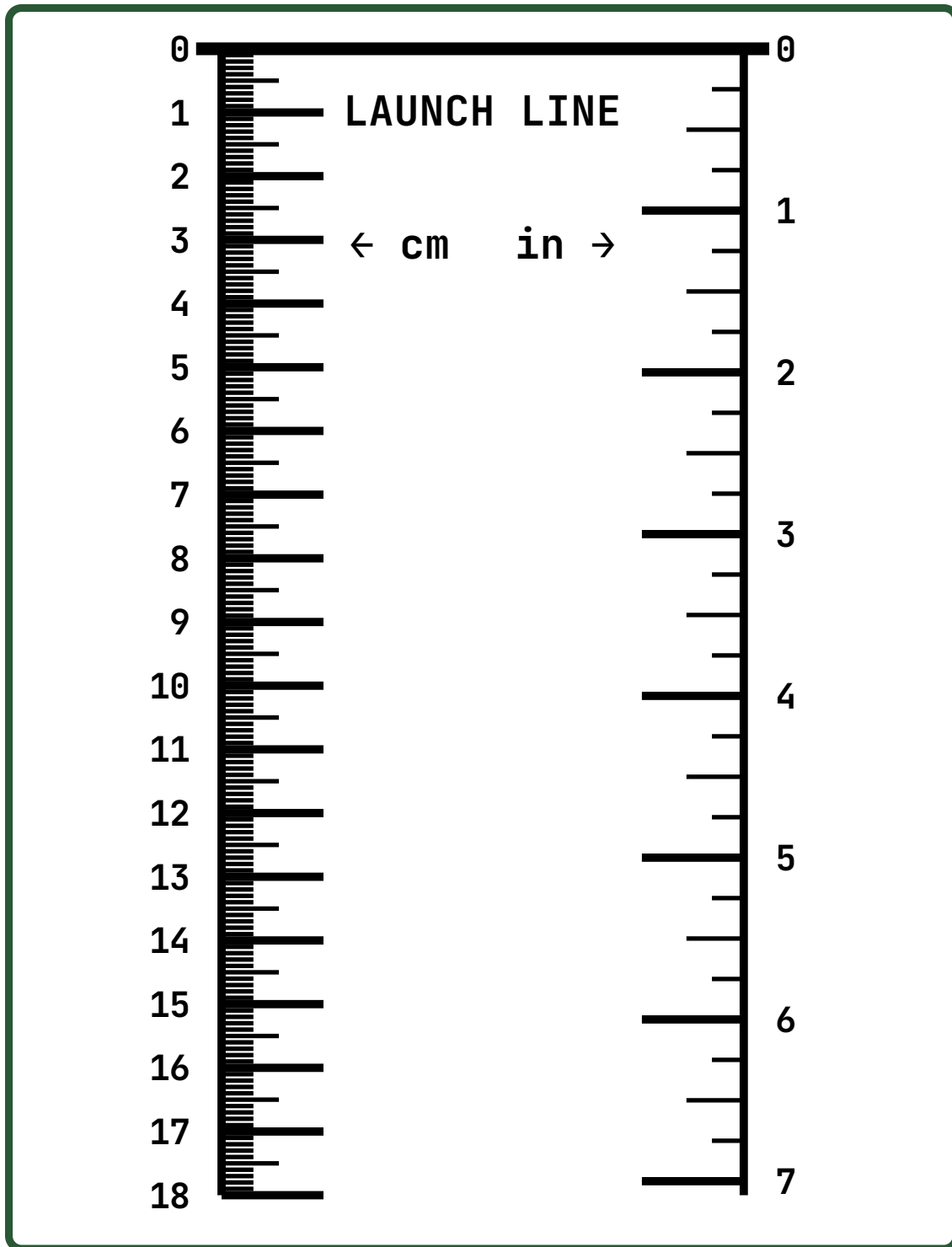
| SCORING CRITERION      | POINTS     |
|------------------------|------------|
| Distance traveled      | 35         |
| Hang time              | 25         |
| Target landing         | 15         |
| Biomimicry explanation | 15         |
| Redesign improvement   | 10         |
| <b>Total</b>           | <b>100</b> |

**SOURCES** Science Buddies Seed Dispersal; Project Learning Tree

## PRINT AND CUT: TTT-03 FLOOR SCALE AND TARGET

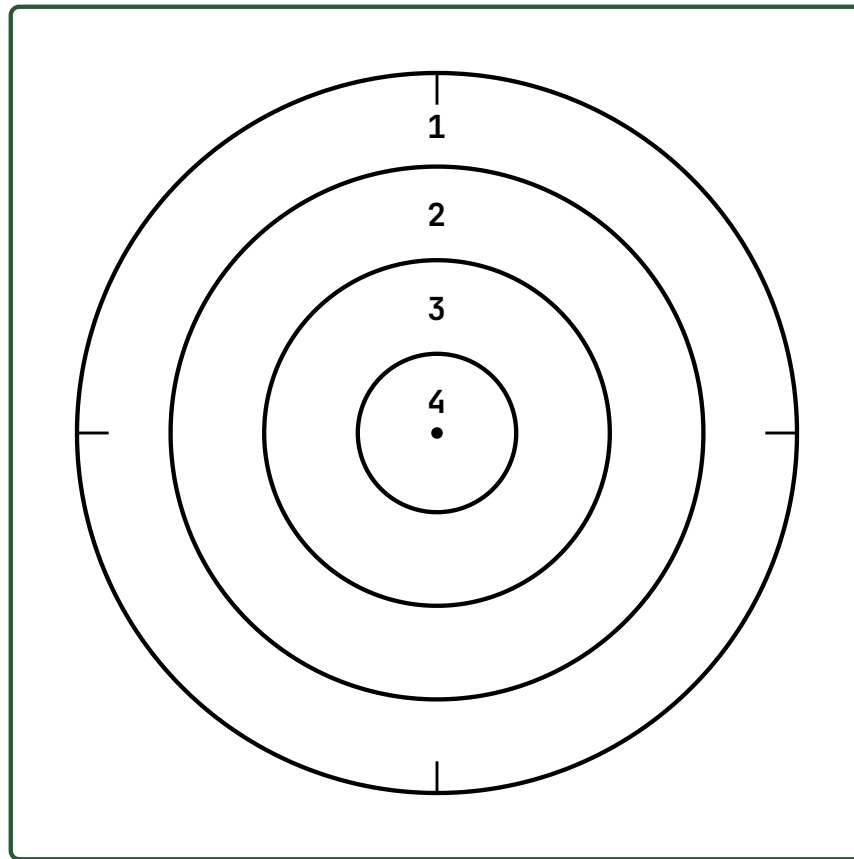
Print at 100% (not "fit to page") on cardstock so the centimeter ruler stays true to size. Tape the strip flat down the lane with the **0** end at the launch line; for a longer lane print several strips and tape them head to tail.

### Drop-lane distance strip



## Landing-zone target

Lay the target flat where staff pick down the lane and line the crosshairs up with the lane center. A higher ring number is a better landing.



### Ring 1

Outer ring. Your seed touched the target.

### Ring 2

Good aim. Getting closer.

### Ring 3

Close to the center.

### Ring 4

Bullseye, right on the dot.

# Xylem Pipeline Relay

Move water uphill the way a tree does

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**At a glance:** Move water uphill like a tree, faster and cleaner than rival teams. About 80 min; 4 teams.

## MATERIALS

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- paper towels
- cotton string
- felt strips
- cups
- food coloring
- trash bag table cover
- rulers
- elevation blocks

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Which material moved water best, and why?
- How did slope change your delivery?
- Where does a real tree use this same effect?

SCORING (100 POINTS)

| SCORING CRITERION  | POINTS |
|--------------------|--------|
| Water delivered    | 40     |
| Speed              | 20     |
| Least spill        | 15     |
| Design explanation | 15     |
| Redesign gain      | 10     |
| Total              | 100    |

SOURCES Science Buddies; plant water transport references

# Greenhouse Climate Controller

Match plants to the right climate zone

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**At a glance:** Run the greenhouse controls for a plant that cannot complain, only wilt. About 80 min; 4 teams.

## MATERIALS

---

- plant profile cards
- control dial boards
- clipboards
- dry erase markers
- tour clue sheets

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?



**SCORING (100 POINTS)**

| SCORING CRITERION       | POINTS     |
|-------------------------|------------|
| Correct setting choices | 30         |
| Use of tour evidence    | 25         |
| Explanation             | 20         |
| Layout efficiency       | 15         |
| Teamwork                | 10         |
| <b>Total</b>            | <b>100</b> |

**SOURCES** Temple Ambler Greenhouse

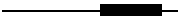
## From Trees to Tech 2026 · TTT-05 Greenhouse Climate Controller

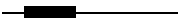
### Plant profile cards

Each card says what one plant needs. Match it to the greenhouse zone whose tour readings fit, then defend the placement with a clue from the tour.

#### Fern

**TEMP**  cool to warm, 16 to 24 °C (60 to 75 °F)

**HUMIDITY**  high, 50% or more; likes damp air

**LIGHT**  shade or bright indirect; never direct sun

**WANTS** Damp air and shade, like a forest floor.

**AVOID** Direct sun and dry air crisp the fronds.

#### Moth orchid

**TEMP**  warm, 24 to 29 °C (75 to 85 °F) by day, above 16 °C (60 °F) at night

**HUMIDITY**  humid, 50 to 80%, with some air movement

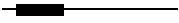
**LIGHT**  medium; bright but filtered, no direct beam

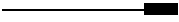
**WANTS** Warm, humid air and bright, filtered light.

**AVOID** Full direct sun scorches the leaves (the orchid-loves-sun myth).

#### Cactus

**TEMP**  warm days, big day-to-night swing, 24 to 34 °C (75 to 90 °F)

**HUMIDITY**  low; dry air and fast-draining soil

**LIGHT**  very high; full, direct sun

**WANTS** Hot, bright, dry air like a desert shelf.

**AVOID** Damp, shady air rots the roots.

#### How to read a card

The thick black bar on each scale is the plant's happy range.

**TEMP**  
10 to 40 °C.

**HUMIDITY**  
0 to 100%.

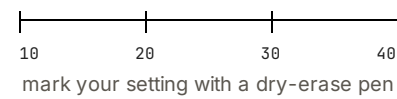
**LIGHT**  
shade on the left to full sun on the right. A plant in the wrong zone tells you how it suffers: too dry wilts, too bright scorches, too damp molds.

From Trees to Tech 2026 · TTT-05 Greenhouse Climate Controller

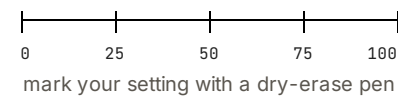
Control dial board

Set each dial for the plant you are placing, then record where each plant goes and the tour clue that backs it.

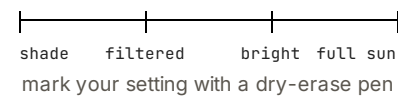
TEMPERATURE (°C)



HUMIDITY (% RH)



LIGHT



Placements: defend each one with a tour clue

| PLANT       | ZONE YOU CHOSE | TEMP | HUMIDITY | LIGHT | TOUR CLUE THAT SUPPORTS IT |
|-------------|----------------|------|----------|-------|----------------------------|
| Fern        |                |      |          |       |                            |
| Moth orchid |                |      |          |       |                            |
| Cactus      |                |      |          |       |                            |
|             |                |      |          |       |                            |
|             |                |      |          |       |                            |

From Trees to Tech 2026 · TTT-05 Greenhouse Climate Controller

Tour clue sheet

As you tour, record what you see, feel, and hear for each zone: warm or cool, damp or dry, bright or shaded, plus any reading the docent points out. Back at the table this is your evidence for matching plants to zones.

| GREENHOUSE ZONE | TEMPERATURE | HUMIDITY<br>(DAMP/DRY) | LIGHT<br>(SHADE/BRIGHT) | WHAT THE DOCENT SAID /<br>EVIDENCE |
|-----------------|-------------|------------------------|-------------------------|------------------------------------|
| Main greenhouse |             |                        |                         |                                    |
| Hoop house      |             |                        |                         |                                    |
| Shaded bench    |             |                        |                         |                                    |
| Bright bench    |             |                        |                         |                                    |
|                 |             |                        |                         |                                    |
|                 |             |                        |                         |                                    |
|                 |             |                        |                         |                                    |

Tip: you do not need exact numbers. "Warmer and more humid than the hoop house" is good evidence. Use the control-system or weather-station readout if the docent shows one.

# Pinecone Weather Machine

A humidity sensor with no battery and no code

---

**At a glance:** Build a humidity sensor with no battery and no code. About 80 min; 4 teams.

## MATERIALS

---

- dry pine cones
- index cards
- wood skewers
- spray bottles
- rulers
- masking tape
- paper fasteners

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Why does a two-layer strip bend when only one layer swells?
- What made your indicator faster?
- Where would a no-power humidity sensor be useful?

SCORING (100 POINTS)

| SCORING CRITERION      | POINTS |
|------------------------|--------|
| Response accuracy      | 35     |
| Response speed         | 20     |
| Readable output scale  | 20     |
| Biomimicry explanation | 15     |
| Redesign quality       | 10     |
| Total                  | 100    |

SOURCES AskNature Pine Cones Strategy

# Pollinator Network Draft and Build

Design a habitat that works all season

---

**At a glance:** Draft a pollinator habitat that works all season, not just on the prettiest day. About 80 min; 4 teams.

## MATERIALS

---

- plant cards
- pollinator cards
- grid boards
- stickers
- markers
- season tokens

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Where was the biggest gap in your bloom calendar?
- Why does clumping help pollinators?
- What happens to the network if one season is missing?

SCORING (100 POINTS)

| SCORING CRITERION         | POINTS |
|---------------------------|--------|
| Seasonal coverage         | 30     |
| Pollinator fit            | 25     |
| Native and clumping logic | 20     |
| Layout clarity            | 15     |
| Defense                   | 10     |
| Total                     | 100    |

SOURCES USDA Pollinators; US Forest Service pollinator guidance



# Arboretum Eco-Quest

Solve outdoor ecology checkpoints

---

**At a glance:** Read the living collection and solve the checkpoints before time runs out. About 80 min; 4 teams.

## MATERIALS

---

- clue cards
- route map
- clipboards
- pencils
- evidence token bags
- optional QR codes

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

**SCORING (100 POINTS)**

| SCORING CRITERION    | POINTS     |
|----------------------|------------|
| Checkpoint accuracy  | 50         |
| Evidence quality     | 20         |
| Pace                 | 20         |
| Teamwork and conduct | 10         |
| <b>Total</b>         | <b>100</b> |

**SOURCES** Temple Ambler Arboretum; Arboretum Explorer

# Minecraft Tree World Resilience Cup

Design a climate-resilient landscape

**At a glance:** Design a landscape that survives stress because of how it is built, not luck. About 80 min; 4 teams.

## MATERIALS

- Minecraft Education devices and template or paper grid kit
- rubric cards
- display timer

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Which feature would help most in a real storm?
- Why does diversity add resilience?
- What in your build would fail first, and how would you fix it?

## SCORING (100 POINTS)

| SCORING CRITERION    | POINTS     |
|----------------------|------------|
| Resilience features  | 30         |
| Biodiversity support | 25         |
| Realism              | 20         |
| Explanation          | 15         |
| Build polish         | 10         |
| <b>Total</b>         | <b>100</b> |

**SOURCES** Minecraft Education Biodiversity

# Tree Ring Climate Detective

Read rings to reconstruct past climate

---

**At a glance:** Read the rings to reconstruct the climate events a tree lived through. About 80 min; 4 teams.

## MATERIALS

---

- printed ring cards
- wood cookies or images
- answer boards
- timers
- claim-evidence cards

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Which pattern was easiest to misread?
- How is a tree ring like and unlike a thermometer?
- What other natural records store past climate?

**SCORING (100 POINTS)**

| SCORING CRITERION      | POINTS     |
|------------------------|------------|
| Correct inferences     | 40         |
| Evidence justification | 20         |
| Speed                  | 20         |
| Final synthesis        | 20         |
| <b>Total</b>           | <b>100</b> |

**SOURCES** NOAA Tree Rings; NOAA Climate.gov; UCAR Center for Science Education

# Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

---

**At a glance:** Build a surface that sheds water and dirt the way a lotus leaf does. About 80 min; 4 teams.

## MATERIALS

---

- wax paper
- cardstock
- sandpaper strips
- masking tape
- pipettes
- water
- pepper or cocoa residue
- ramps

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

SCORING (100 POINTS)

| SCORING CRITERION          | POINTS |
|----------------------------|--------|
| Fast clean runoff          | 35     |
| Least residue              | 25     |
| Surface design explanation | 20     |
| Redesign gain              | 10     |
| Craftsmanship              | 10     |
| Total                      | 100    |

SOURCES AskNature Lotus Leaf Self-Cleaning Surface

# Leaf Stomata Microscope Detective

Count stomata and rank leaves by water strategy

---

**At a glance:** Count the breathing pores and rank leaves by how they manage water. About 80 min; 4 teams.

## MATERIALS

---

- leaf samples
- clear tape
- clear nail polish or prepared slides
- USB microscopes or magnifiers
- grid sheets
- gloves

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

---

- Why count several fields instead of one?
- What does a high stomata count suggest about a leaf's habitat?
- Where might error sneak into your count?



## SCORING (100 POINTS)

| SCORING CRITERION     | POINTS     |
|-----------------------|------------|
| Slide or peel quality | 25         |
| Counting accuracy     | 30         |
| Ranking evidence      | 25         |
| Team data table       | 10         |
| Cleanup and care      | 10         |
| <b>Total</b>          | <b>100</b> |

**SOURCES** Leaf stomata microscopy adaptations; Science Buddies plant biology

## BACKUP ACTIVITIES

# Ready alternates

# Root Grip Erosion Rescue

Can roots hold a slope together in a storm?

**At a glance:** Can roots hold a slope together during a storm? About 60 min; 4 teams.

## MATERIALS

- Trays
- Soil
- String or yarn roots
- Craft sticks
- Watering cups

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why did the rooted slope hold more soil?
- How would deeper roots change the result?
- Where is this used to protect real hillsides?

## SCORING (100 POINTS)

| SCORING CRITERION  | POINTS     |
|--------------------|------------|
| Soil retained      | 40         |
| Runoff control     | 25         |
| Design explanation | 20         |
| Cleanup            | 15         |
| <b>Total</b>       | <b>100</b> |

**SOURCES** EPA green infrastructure and erosion lessons

# Tree Height Triangulation Shootout

Measure a tree without climbing it

**At a glance:** Measure a tree without climbing it. About 60 min; 4 teams.

## MATERIALS

- Paper clinometers
- String and washers
- Rulers and tape measures
- Clipboards

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- What was your biggest source of error?
- Why does eye height matter in the calculation?
- Where is remote height measurement used in the field?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS     |
|-------------------|------------|
| Accuracy          | 50         |
| Method clarity    | 20         |
| Speed             | 20         |
| Teamwork          | 10         |
| <b>Total</b>      | <b>100</b> |

SOURCES GLOBE Trees Activities

# Urban Heat Shade Dash

Find the coolest route through a hot campus

**At a glance:** Find the coolest route through a hot campus. About 60 min; 4 teams.

## MATERIALS

- Thermometers or IR thermometers
- Campus maps
- Clipboards

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- How big was the sun-to-shade temperature gap?
- What surface was hottest, and why?
- How do cities use shade to cool streets?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS     |
|-------------------|------------|
| Cool route design | 35         |
| Data quality      | 25         |
| Reasoning         | 20         |
| Communication     | 20         |
| <b>Total</b>      | <b>100</b> |

**SOURCES** NASA MyNASADData Urban Heat Island

# Photosynthesis Float-Off Playoffs

Make leaf disks float by making oxygen

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**At a glance:** Make leaf disks float by producing oxygen. About 60 min; 4 teams.

## MATERIALS

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- Spinach leaves
- Baking soda solution
- Cups
- Oral syringes, no needle
- Timers

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why does oxygen make the disks float?
- Which variable changed the rate most?
- How is this evidence of photosynthesis?

## SCORING (100 POINTS)

| SCORING CRITERION          | POINTS     |
|----------------------------|------------|
| Performance                | 35         |
| Controlled-variable design | 25         |
| Data table                 | 20         |
| Explanation                | 10         |
| Cleanup                    | 10         |
| <b>Total</b>               | <b>100</b> |

**SOURCES** Exploratorium Photosynthetic Floatation